

Grzegorz Mróz

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Computer Science master's student (Machine Learning & AI, active student status) with a BSc in Mathematics and hands-on experience building LLM-based solutions in Python - RAG pipelines, embeddings and semantic search, prompt engineering and output evaluation (Anthropic API, LangChain). Professional background in data engineering and quantitative modelling at UBS. Eager to build and iterate on practical GenAI prototypes for real business use cases.

EXPERIENCE

- **UBS** Kraków, Poland
Quantitative Analyst, Risk July 2026 - Present
 - **Quantitative Modelling:** Develop quantitative risk models within the Pillar 2 framework, implementing and analysing models in Python and SQL.
- **UBS** Kraków, Poland
Data Analyst August 2025 - February 2026
 - **Data Pipelines:** Built and maintained production Python (pandas, NumPy) and SQL data pipelines, collecting, cleaning and structuring large datasets for critical regulatory reporting with strict data integrity.
 - **Data Quality & Troubleshooting:** Designed statistical data-quality checks and diagnosed root causes of data incidents with cross-functional engineering teams (Agile, Jira, Git), improving reliability of downstream reporting.

SELECTED PROJECTS

- **RAG Document Assistant: Grounded LLM Q&A with Citations**
Python, LangChain, Anthropic API, Chroma, AWS
 - **RAG Pipeline:** Built an end-to-end RAG pipeline in Python (LangChain): document ingestion and chunking, embedding-based semantic search over a Chroma vector database, and grounded LLM answer generation with source citations via the Anthropic API.
 - **Evaluation:** Designed prompts and an evaluation loop (pandas / NumPy) scoring answer groundedness against retrieved context and flagging hallucinations across a test-question set.
 - **Deployment:** Served the assistant as a REST API (FastAPI) with a lightweight UI; Dockerised and deployed on AWS (EC2), with source versioned on GitHub.
- **World Models & Model-Based RL: Dreamer-style Agent**
Python, PyTorch
 - **Model-Based RL:** Implemented a Recurrent State-Space Model (RSSM) neural world model (deterministic GRU path + stochastic latents) learning environment dynamics and rewards, then trained an actor-critic policy entirely in imagined latent rollouts.
 - **Benchmarking:** Benchmarked sample efficiency against model-free reinforcement learning baselines (PPO / SAC / DQN) on pixel-based control tasks, with latent reconstruction visualisations and imagination-horizon ablations.

EDUCATION

- **AGH University of Krakow** Kraków, Poland
Master of Engineering in Computer Science, Machine Learning and AI October 2025 - 2027 (Expected)
 - **Core modules:** Machine Learning Algorithms, Neural Networks, RL-DAS, Python, Java, Computing intelligence, Statistics, Artificial Intelligence, Reinforcement Learning.
- **AGH University of Krakow** Kraków, Poland
Bachelor of Science in Mathematics October 2022 - July 2025
 - **Foundations:** Probability Theory, Statistics, Linear Algebra, Differential Equations, Introduction to Numerical Analysis.

TECHNICAL SKILLS

- **GenAI & LLMs:** LLM integration (Anthropic API, REST APIs), Claude Code, LangChain, RAG, embeddings & semantic search, vector databases, prompt engineering, output evaluation & hallucination detection.
- **Programming:** Python (pandas, NumPy, OOP, REST APIs), SQL, Java; Git/GitHub
- **Machine Learning & AI:** PyTorch, scikit-learn; neural networks; reinforcement learning, world models).
- **Data, Cloud & Tools:** AWS basics (EC2, S3), Docker, Jira, Jupyter.
- **Mathematics:** Linear algebra, probability theory, probability & statistics.

AWARDS & LANGUAGES

- **HackYeah 2025 Finalist:** Europe's largest stationary hackathon; delivered a working prototype in a team under time pressure.

Languages: Polish (Native), English (C1)

I agree to the processing of my personal data for the purposes of the recruitment process, in accordance with Regulation (EU) 2016/679 (GDPR) and the Personal Data Protection Act of 10 May 2018 (Journal of Laws 2018, item 1000).